

Exam Date & Time: 16-Jan-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - JANUARY-2024

Computer Graphics and Animations [IT-4301]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Construct the line from $(-1,1)$ to $(5,-8)$ using Bresenham's line drawing Algorithm. (5)
 - A)
 - B) Identify the significance of OpenGL library functions. (3)
 - C) Identify the matrix representation for scaling, translation, and rotation (2)
- 2) Let ABCD be the regular window with $A(20, 20)$, $B(90, 20)$, $C(90, 70)$, and $D(20, 70)$. Examine the region codes for endpoints and use the Cohen-Sutherland algorithm to clip the lines $P(10, 30)$ and $Q(80, 90)$. (5)
 - A)
 - B) Identify the possible four cases in Sutherland-Hodgman Polygon Clipping. (5)
- 3) Infer the transformation matrix for 3D scaling followed by rotation about a fixed point (5)
 - A)
 - B) Identify the procedure to transform the scene from the world coordinate system to the viewing coordinate system in 3D viewing. (5)
- 4) Develop a raster transformation for 2D objects using OpenGL. (5)
 - A)
 - B) Analyze the matrix for the rotation of an object about an arbitrary axis. (5)
- 5) Utilize the Scan-Line Polygon Fill Algorithm for a 2D object. (5)
 - A)
 - B) Identify the tricks used in computer graphics animation (3)
 - C) Develop a code to draw a line in OpenGL. (2)

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